Tema-Probleme ramase-seminar 3

1. Ordonarea crescatoarea a 3 nr. reale.

```
{ read a,b,c,aux;
  { if a>b then
  aux=a;
  a=b;
  b=aux;
endif;
  }
    { if a>c then
  aux=a;
  a=c;
  c=aux;
endif;
  }
  { if b>c then
  aux=b;
  b=c;
  c=aux;
endif;
  }
  write a,b,c;
 }
```

3. Sa se resolve ecuatia unde a,b sunt numere reale: ax2+bx+c=0.

```
 { read a,b,c,x1,x2;
```

```
  { if a!=0 then
  x1=((-b+sqrt(b*b-4*a*c)/2*a);
  x2=((-b-sqrt(b*b-4*a*c)/2*a);
  else
  { if a=0 then
  x1=-c/b;
  x2=-c/b;
 else
  if b=0 then
  write "x poate fi orice nr.";
else
  write "nu are solitii nenule";
endif;}
endif;}
  write x1,x2;
}
```

Probleme ramase cu vectori-seminar 4

7. Elementul maxim intr-un vector si pozitiile tuturor aparitiilor sale.

```
{ read n,(x(i),i=1,n), max;
  max=x(1);
do for i=2,n,1;
 if x(i)>max then
  max=x(i);
endif;
enddo;
 do for i=1,n,1
```

```
if max=x(i) then
  write i;
  write " ";
endif;
enddo;
}
```

8. Determinati amplitudinea elementelor intr-un vector.

```
 { read n,(x(i),i=1,n),A;
  max=x(1);
  do for i=2,n,1
  if x(i)>max then
  max=x(i);
endif;
enddo;
  min=x(1);
    do for i=2,n,1
  if x(i)<min then
  min=x(i);
endif;
enddo;
  A=max-min;
  write A;
}
```

9. Calculati valoarea unui polinom intr-un punct dat.

```
{ read n,(x(i),i=1,n),polinom,p;
  polinom=x(1);
```

```
  p=1;
  do for i=2,n,1
  { p=p*x;
  polinom=polinom+x(i)*p;
}
endif;
enddo;
  write polinom;
}
```

10. Sa se insereze o valoare data intr-un vector ordonat de n elemente astfel incat dupa inserare vectorul sa ramana ordonat.

i) Daca vectorul e ordonat crescator:

```
 { read n,(x(i),i=1,n),c,j;
 do for i=1,n,1
  { if c<v(i) then
  { do for j=n,i+1,-1
  { v(j)=v(j-1);
  n=n+1;
  i=i+1;
enddo;}
endif;}
enddo;}
  write (v(j),j=1,n);
}
```

ii) Daca vectorul e ordonat descrescator:

```
 { read n,(x(i),i=1,n),c,j;
```

```
 do for i=1,n,1
  { if c>v(i) then
  { do for j=n,i+1,-1
  { v(j)=v(j-1);
  n=n+1;
  i=i+1;
enddo;}
endif;}
enddo;}
  write (v(j),j=1,n);
}
```